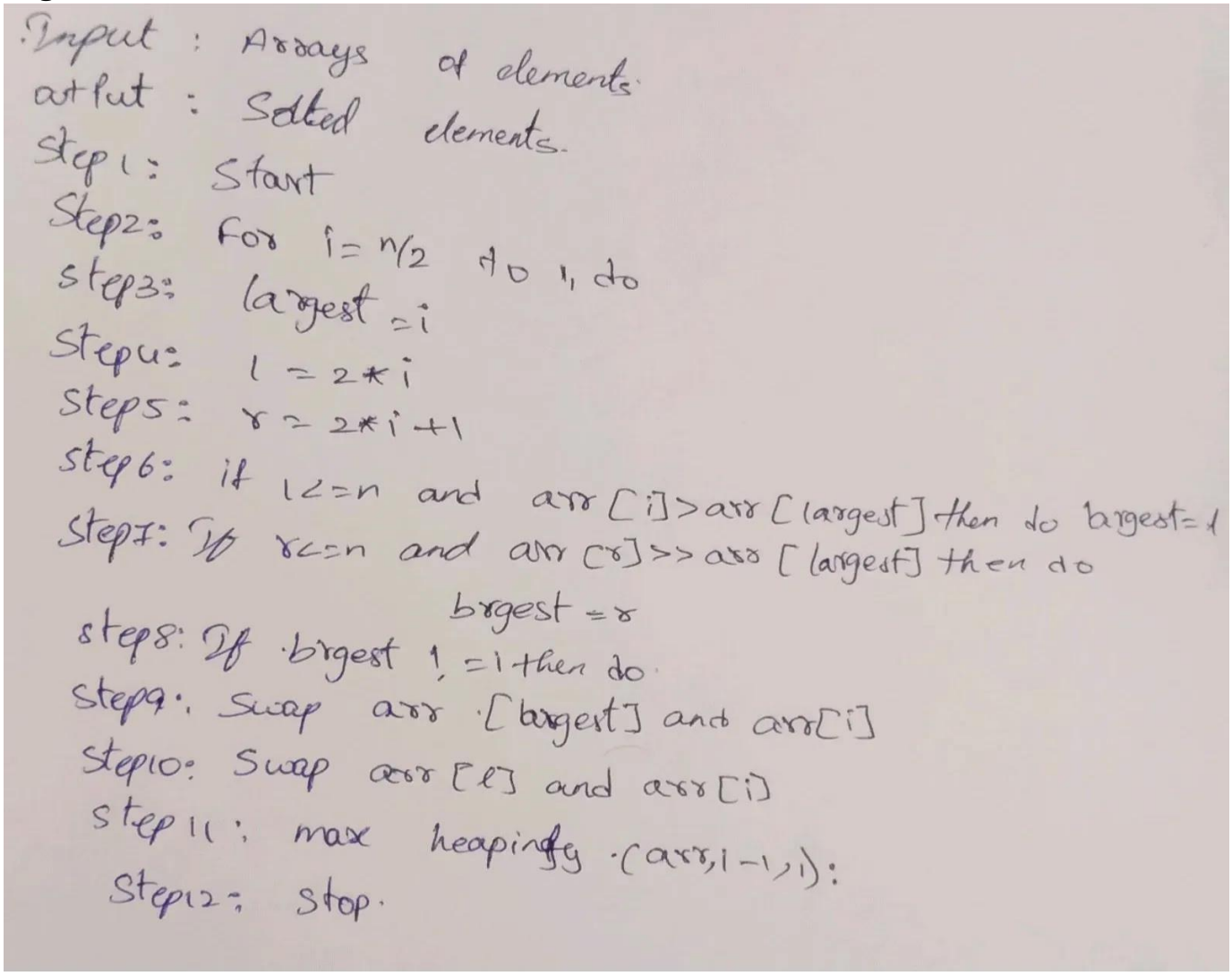
Practical: 3	**Implementation of max-heap sort algorithm.**
11/08/2026	

Aim: To Implement of max-heap sort algorithms.

MAX-H EAP SORT :

Algorithm:



Program:

```
#include <iostream>
using namespace std;
 void heapify(int arr[], int n, int i) {
  int largest = i;
  int left = 2 * i + 1;
int right = 2 * i + 2;
 if (left < n && arr[left] > arr[largest]) {
largest = left;
    }     if (right < n && arr[right] >
arr[largest]) {         largest = right;
    }     if (largest != i) {
swap(arr[i], arr[largest]);
heapify(arr, n, largest);
    }
} void heapSort(int arr[], int n) {
for (int i = n / 2 - 1; i >= 0; i--) {
heapify(arr, n, i);
    }     for (int i = n - 1; i > 0;
i--) {         swap(arr[0], arr[i]);
heapify(arr, i, 0);
    }
} int main() {     int arr[] =
{4, 10, 3, 5, 1};     int n = 5;
heapSort(arr, n);     for (int i
= 0; i < n; i++) {         cout
<< arr[i] << " ";
    }
return 0;
}
```

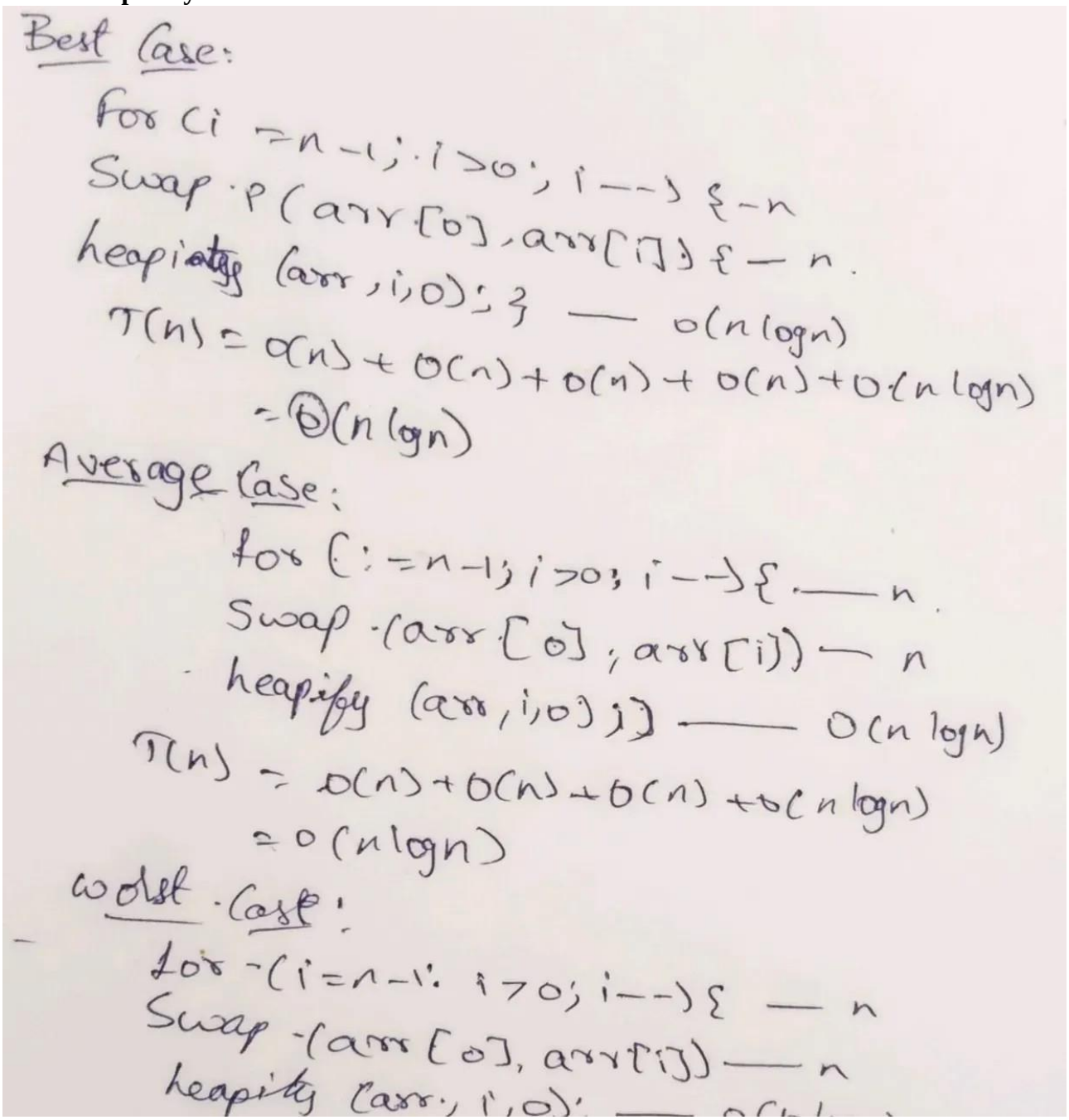
Output:

```
Output

Enter the number of elements: 5
Enter 5 elements:
3
1
6
4
2
Array after Heap Sort: 1 2 3 4 6

=== Code Execution Successful ===
```

Time Complexity:



Conclusion:

The implementation and time analysis of Max-Heap Sort helps us understand its sortingefficiency. Max-Heap Sort builds a Max Heap and repeatedly swaps the maximum element withthe last element to sort the array in ascending order. It has a time complexity of O(n log n) in thebest, average, and worst cases. Therefore, Max-Heap Sort is an efficient and consistent sortingalgorithm for handling large datasets.